

Challenge (Habanero)**Q1.** What is the expansion of $(x + 3)^2$?

- (A) $x^2 + 6x + 9$
 (B) $x^2 - 6x + 9$
 (C) $x^2 + 3x + 9$
 (D) $x^2 - 3x + 9$
 (E) $x^2 + 9$

Q2. A gaming company uses the expression $x^2 - 10x + 25$ to model the score of a new game. What is the factored form of this expression?

- (A) $(x - 5)^2$
 (B) $(x + 5)^2$
 (C) $(x - 2)(x - 5)$
 (D) $(x + 2)(x + 5)$
 (E) $(x - 1)(x - 10)$

Q3. A coder is debugging a line of code that contains the expression $(2x + 5)^2$. What is the expanded form of this expression?

- (A) $4x^2 + 20x + 25$
 (B) $4x^2 - 20x + 25$
 (C) $2x^2 + 10x + 25$
 (D) $2x^2 - 10x + 25$
 (E) $4x^2 + 10x + 25$

Q4. A data analyst is working with the expression $9x^2 - 12x + 4$. What type of trinomial is this?

- (A) Perfect square trinomial
 (B) Difference of squares
 (C) Sum of cubes
 (D) Difference of cubes
 (E) None of the above

Q5. What is the factored form of the expression $16x^2 - 24x + 9$?

- (A) $(4x - 3)^2$
 (B) $(4x + 3)^2$
 (C) $(2x - 3)(2x - 3)$
 (D) $(2x + 3)(2x + 3)$
 (E) $(4x - 1)(4x - 9)$

Q6. A gaming platform uses the expression $(3x - 2)^2$ to model the number of players online. What is the expanded form of this expression?

- (A) $9x^2 - 12x + 4$
 (B) $9x^2 + 12x + 4$
 (C) $9x^2 - 6x + 4$
 (D) $9x^2 + 6x + 4$
 (E) $9x^2 - 4x + 4$

Q7. What is the expansion of $(2x + 1)^2$?

- (A) $4x^2 + 4x + 1$
 (B) $4x^2 - 4x + 1$
 (C) $2x^2 + 4x + 1$
 (D) $2x^2 - 4x + 1$
 (E) $4x^2 + 1$

Q8. A software company uses the expression $x^2 + 14x + 49$ to model the number of users. What is the factored form of this expression?

- (A) $(x + 7)^2$
 (B) $(x - 7)^2$
 (C) $(x + 1)(x + 49)$
 (D) $(x - 1)(x - 49)$
 (E) $(x + 2)(x + 24)$

Open Ended Questions (FRQ)**Q9.** Factor the expression $25x^2 - 30x + 9$. Show all work and explain your reasoning.**Q10.** Expand the expression $(x + 2)(x - 5)$ and then factor the result. Explain the relationship between the original expression and the factored form.**Chef's Tips**

- Remind students to simplify their answers
- Encourage students to check their work
- Emphasize the importance of showing all work and explaining reasoning